

Smart City Bike Sharing — Network Performance Analysis

Data Analytics Module-End Project Using Power Query, Power BI, DAX

1. Project Overview and Objective

This project analyzes public bike-sharing station data spanning 25 city contracts to evaluate station capacity, live bike availability, operational status, banking facilities, bonus-enabled stations, and update patterns over time. The dataset was cleaned and transformed in Power Query, structured into a star-schema data model in Power BI, and analyzed using 16 custom DAX measures.

Objectives:

- Analyze bike station distribution across 25 city contracts.
- Monitor bike availability and empty docking stands across the network.
- Identify operational (OPEN) versus inactive (CLOSED) stations.
- Compare banking-enabled and non-banking stations.
- Analyze bonus-enabled stations.
- Evaluate how station data updates are distributed across 2022-2025.
- Build an interactive Power BI dashboard for operational monitoring.

2. Data Source

Dataset	Bike Sharing Station Dataset
Domain	Urban Transportation / Bike Sharing
Data Type	Operational station-level snapshot data
Geographic Coverage	25 city contracts across multiple countries
File Format	Excel (.xlsx)
Tools Used	Excel, Power Query, Power BI (DAX)
Total Records	2,856 rows

3. Problem Statement

Bike-sharing systems generate continuous operational data from hundreds of stations. Efficient monitoring of bike availability, station capacity, and operational status is essential for reliable service. This project answers:

- How many bike stations exist, and how much total capacity do they provide?
- How many bike stands and bikes are currently available?
- Which stations/contracts are currently operational versus closed?
- Which stations support banking facilities, and how many offer bonus incentives?
- Which contracts (cities) have the highest station capacity?
- How is station update data distributed across 2022-2025?
- Are there inconsistencies between total bike stands and reported bike/stand availability?

4. Dataset Attributes

Attribute	Description
number	Unique station identifier
name / Station Address	Bike station name and address
position → Latitude, Longitude	Split into two numeric fields for mapping
banking	Whether the station has a card payment terminal (TRUE/FALSE)
bonus	Whether the station is a bonus-incentive station (TRUE/FALSE)
status	OPEN or CLOSED
Bike Stands	Total docking capacity of the station
Available Bike Stands	Empty docks, ready to receive a returned bike
Available Bikes	Bikes currently available to rent
Last Update / Update Month Year	Timestamp of last refresh, and derived month-year label
Contract Name	City / network operator (25 unique contracts)

5. Tools and Technologies

Power Query

- Data cleaning: null handling, text trimming/cleaning
- Data type conversion (text to number, text to date)
- Splitting position into Latitude and Longitude
- Building Dim_City and Dim_Date reference tables

Power BI

- Star-schema data modeling with fact and dimension tables
- DAX calculated columns and measures
- Interactive dashboard: KPI cards, charts, map, slicers

6. Data Cleaning and Transformation (Power Query)

The following cleaning steps were performed before modeling:

- Checked and confirmed no null values remained in key columns after cleaning.
- Checked for duplicate records using both full-row and composite-key (Station ID + Contract Name) comparisons — none found.
- Replaced blank Address values with "Test Station" to correctly label internal test entries
- Converted Available Bikes, Bike Stands, and Available Bike Stands from Text to numeric types where needed for DAX calculations.
- Split the combined position field into separate Latitude and Longitude numeric columns for map visualization.
- Extracted a Month-Year text label from the Last Update timestamp for time-based analysis.

Station ID	Station ID - Street Label	Station Address	position	banking	bonus	status	Contract
5	005_GLORIETA DE LOS DELFINES	Glorieta de los Delfines (Aprox. Avda de los Castros)	43.4719661074837, -3.79285449619735	TRUE	FALSE	CLOSED	Santander
2	02- PASTEUR - FAC DE DROIT	AVENUE PASTEUR	49.4442849736749, 1.07838302560908	TRUE	FALSE	CLOSED	Rouen
5	05- HOTEL DE VILLE	RUE DE L HOPITAL	49.44250558644192, 1.098266338219107	TRUE	FALSE	CLOSED	Rouen
22	22 - CATHEDRALE	Rue des Carmes angle rue du Gros Horloge	49.440904, 1.09396	TRUE	FALSE	CLOSED	Rouen
16	016_SEVERO OCHOA	Severo Ochoa	43.4528130549523, -3.87139068298249	TRUE	FALSE	OPEN	Santander
50013	50013 - CER13 - ROND-POINT DES RAIES	ALLÉE DU BELVÈDÈRE ANGLE ROND POINT DES RAIES - 95000 CERGY	49.044831, 2.053462	FALSE	FALSE	CLOSED	Cergy-Po
50022	50022 - CER22 - GARE CERGY GRAND CENTRE (CB)	BOULEVARD DE L'OISE / RUE DE LA GARE - 95000 CERGY	49.036655, 2.080189	TRUE	FALSE	OPEN	Cergy-Po
50005	50005 - CER05 - MOULIN A VENT	AVENUE DU HAZAY ANGLE RUE DE L'AISSLETTE - 95000 CERGY	49.049401, 2.017075	FALSE	FALSE	CLOSED	Cergy-Po
53002	53002 - PON02 - PLACE DE LA LIBÉRATION	Place de la Libération côté avenue de Verdun - 95300 PONTOISE	49.044395, 2.090632	FALSE	FALSE	CLOSED	Cergy-Po
50014	50014 - CER14 - AVENUE DU NORD	AVENUE DU NORD ANGLE ALLÉE DES ACACIAS - 95000 CERGY	49.043224, 2.060405	FALSE	FALSE	CLOSED	Cergy-Po
241	241_PLAZA NUEVA	PLAZA NUEVA - Aprox. C/ Mendez Núñez	37.388882, -5.996237	FALSE	FALSE	OPEN	Seville
4	04- MUSEE DES BEAUX ARTS	DEVANT N° 39 ALLEE EUGENE DELACROIX	49.44430797679112, 1.093579586130271	TRUE	FALSE	CLOSED	Rouen
4	004_CORMORAN	Cormoran (Aprox. C/ de Manuel García Lago)	43.4784718110308, -3.78865314567001	TRUE	FALSE	CLOSED	Santander
16	16-ILE LACROIX	AVENUE JACQUES CHASTELAIN - PARKING DE LA PATINOIRE	49.431521256315975, 1.102925918082832	TRUE	FALSE	CLOSED	Rouen
53003	53003 - PON03 - UNIVERSITÉ ST MARTIN	Avenue François Mitterrand côté boulevard du Port - 95300 PONTOISE	49.0435829707344, 2.08480803731778	FALSE	FALSE	CLOSED	Cergy-Po
21	21- MAISON DE LA CULTURE	Place Léon Gontier	49.89392165574068, 2.292724729098195	FALSE	FALSE	OPEN	Amiens
50027	50027 - CER27 - MARIOBERTS	7 Rue des Chauffours, 95000 Cergy	49.041233, 2.075956	FALSE	FALSE	CLOSED	Cergy-Po
50023	50023 - CER23 - LES TOULEUSES	AVENUE DU SUD - 95000 CERGY	49.030665, 2.083067	FALSE	FALSE	CLOSED	Cergy-Po
17	17- CIRQUE	Mail Albert 1er	49.8887, 2.2973	FALSE	FALSE	OPEN	Amiens
2012	2012-GRAND RUE	GRAND RUE - FACE AU 35 GRAND RUE	43.2978095560818, 5.37083509372786	TRUE	FALSE	CLOSED	Marseille
11	11- CLEMENCEAU JOFFRE	COURS CLEMENCEAU	49.43464613956876, 1.087959490140167	TRUE	FALSE	CLOSED	Rouen
2	002_PLAZA DE LAS BRISAS	Plaza de las Brisas (Aprox. Jardines de Piquio)	43.4744126704703, -3.78578134528458	TRUE	FALSE	CLOSED	Santander
3	03- VOGEL	Bd du jardin des plantes	49.898124385191046, 2.295195128091815	FALSE	FALSE	OPEN	Amiens
12	12- PLACE DES COTONNIERS	PLACE DES COTONNIERS	49.43087671678897, 1.081412105910482	TRUE	FALSE	OPEN	Rouen
28	28 - BOULINGRIN	Test Station	49.446882, 1.106504	TRUE	FALSE	CLOSED	Rouen
53004	53004 - PON04 - LA PALETTE	BD DE L'OISE ANGLE RUE DE LA PALETTE	49.032140952207, 2.08780221085683	FALSE	FALSE	CLOSED	Cergy-Po
242	242_PLAZA NUEVA	PLAZA NUEVA - Aprox. C/ Mendez Núñez	37.388844, -5.996232	FALSE	FALSE	OPEN	Seville

Smart City Bike Sharing Data - Last saved: today at 3:45 PM

Search

Sign in

FileHomeHelpTable toolsColumn tools

NameStation Fill Status

Data typeWhole number

Format\$ - %

Summarization

Data category

Sort by columnSort

Data groups

Manage relationships

New columnCalculations

Structure

Formatting

Properties

Sort

Groups

Relationships

Calculations

1 Station Fill Status =

2 IF([Sales[Available Bikes]] = 0, "Empty",

3 IF([Sales[Available Bikes]] = "Sales [Bike Stands]", "Full", "Partial"))

4 DAX comparison operations do not support comparing values of type Text with values of type Integer. Consider using the VALUE or FORMAT function to convert one of the values.

	banking	bonus	status	Contract Name	Bike Stands	Available Bike Stands	Available Bikes	Last Update	Latitude	Longitude	Station Fill Status
	TRUE	FALSE	OPEN	Rouen	20	20	0	30-06-2022 14:08:00 +05:30	49.430876716789	1.08141210591048	#ERROR
	FALSE	FALSE	OPEN	Amiens	20	20	0	28-06-2025 13:50:59 +05:30	49.8887	2.2973	#ERROR
	FALSE	FALSE	OPEN	Seville	20	20	0	23-10-2025 10:05:07 +05:30	37.388844	-5.996232	#ERROR
	FALSE	FALSE	OPEN	Seville	20	20	0	23-10-2025 10:04:38 +05:30	37.388882	-5.996237	#ERROR
	TRUE	FALSE	CLOSED	Marseille	20	20	0	19-12-2022 20:02:52 +05:30	43.2978095560818	5.37083509372786	#ERROR
	TRUE	FALSE	CLOSED	Santander	20	20	0	18-12-2023 17:04:43 +05:30	43.4528130549523	-3.87139068298249	#ERROR
	TRUE	FALSE	CLOSED	Santander	20	20	0	17-01-2024 14:50:00 +05:30	43.4719661074837	-3.79285449619735	#ERROR
	TRUE	FALSE	CLOSED	Santander	20	20	0	17-01-2024 14:39:26 +05:30	43.4784718110308	-3.78865314567001	#ERROR
	TRUE	FALSE	CLOSED	Santander	20	20	0	17-01-2024 14:39:18 +05:30	43.4744126704703	-3.78578134528458	#ERROR
	FALSE	FALSE	CLOSED	Cergy-Pontoise	20	20	0	13-06-2025 12:42:21 +05:30	49.030665	2.083067	#ERROR
	FALSE	FALSE	OPEN	Luxembourg	20	20	0	12-11-2025 14:57:33 +05:30	49.61349	6.17066	#ERROR
	FALSE	FALSE	OPEN	Valencia	20	20	0	12-11-2025 14:57:29 +05:30	39.4845091367472	-0.394769162326153	#ERROR
	FALSE	FALSE	OPEN	Ljubljana	20	20	0	12-11-2025 14:57:26 +05:30	46.06005	14.51302	#ERROR
	FALSE	FALSE	OPEN	Lyon	20	20	0	12-11-2025 14:57:09 +05:30	45.785465	4.847838	#ERROR
	TRUE	FALSE	OPEN	Toulouse	20	20	0	12-11-2025 14:57:05 +05:30	43.600361	1.464039	#ERROR
	FALSE	FALSE	OPEN	Seville	20	20	0	12-11-2025 14:57:03 +05:30	37.397872	-5.976108	#ERROR
	FALSE	FALSE	OPEN	Seville	20	20	0	12-11-2025 14:56:47 +05:30	37.376585	-5.972616	#ERROR
	FALSE	FALSE	OPEN	Luxembourg	20	20	0	12-11-2025 14:56:32 +05:30	49.623837	6.180642	#ERROR
	FALSE	FALSE	OPEN	Seville	20	20	0	12-11-2025 14:55:54 +05:30	37.413162	-5.985356	#ERROR
	FALSE	FALSE	OPEN	Seville	20	20	0	12-11-2025 14:55:44 +05:30	37.40062	-5.986235	#ERROR
	FALSE	FALSE	OPEN	Luxembourg	20	20	0	12-11-2025 14:55:22 +05:30	49.594008	6.125123	#ERROR
	FALSE	FALSE	OPEN	Valencia	20	20	0	12-11-2025 14:55:17 +05:30	39.485943149532	-0.37003608893869	#ERROR
	FALSE	FALSE	OPEN	Manibar	20	20	0	12-11-2025 14:55:11 +05:30	46.56566	15.652935	#ERROR

Data

Search

Sales

Available Bike Stands

Available Bikes

banking

Bike Stands

bonus

Contract Name

Last Update

Latitude

Longitude

Station Address

Station Fill Status

Station ID

Station ID - Street Label

status

Last Update	Latitude	Longitude	Station Fill Status	Update Month Year
30-06-2022 14:08:00 +05:30	49.430876716789	1.08141210591048	Empty	30-06-2022 14:08:00 +05:30
28-06-2025 13:50:59 +05:30	49.8887	2.2973	Empty	28-06-2025 13:50:59 +05:30
23-10-2025 10:05:07 +05:30	37.388844	-5.996232	Empty	23-10-2025 10:05:07 +05:30
23-10-2025 10:04:38 +05:30	37.388882	-5.996237	Empty	23-10-2025 10:04:38 +05:30
19-12-2022 20:02:52 +05:30	43.2978095560818	5.37083509372786	Empty	19-12-2022 20:02:52 +05:30
18-12-2023 17:04:43 +05:30	43.4528130549523	-3.87139068298249	Empty	18-12-2023 17:04:43 +05:30
17-01-2024 14:50:00 +05:30	43.4719661074837	-3.79285449619735	Empty	17-01-2024 14:50:00 +05:30
17-01-2024 14:39:26 +05:30	43.4784718110308	-3.78865314567001	Empty	17-01-2024 14:39:26 +05:30
17-01-2024 14:39:18 +05:30	43.4744126704703	-3.78578134528458	Empty	17-01-2024 14:39:18 +05:30
13-06-2025 12:42:21 +05:30	49.030665	2.083067	Empty	13-06-2025 12:42:21 +05:30
12-11-2025 14:57:33 +05:30	49.61349	6.17066	Empty	12-11-2025 14:57:33 +05:30
12-11-2025 14:57:29 +05:30	39.4845091367472	-0.394769162326153	Empty	12-11-2025 14:57:29 +05:30
12-11-2025 14:57:26 +05:30	46.06005	14.51302	Empty	12-11-2025 14:57:26 +05:30
12-11-2025 14:57:09 +05:30	45.785465	4.847838	Empty	12-11-2025 14:57:09 +05:30
12-11-2025 14:57:05 +05:30	43.600361	1.464039	Empty	12-11-2025 14:57:05 +05:30
12-11-2025 14:57:03 +05:30	37.397872	-5.976108	Empty	12-11-2025 14:57:03 +05:30
12-11-2025 14:56:47 +05:30	37.376585	-5.972616	Empty	12-11-2025 14:56:47 +05:30
12-11-2025 14:56:32 +05:30	49.623837	6.180642	Empty	12-11-2025 14:56:32 +05:30

7. Data Modeling (Star Schema)

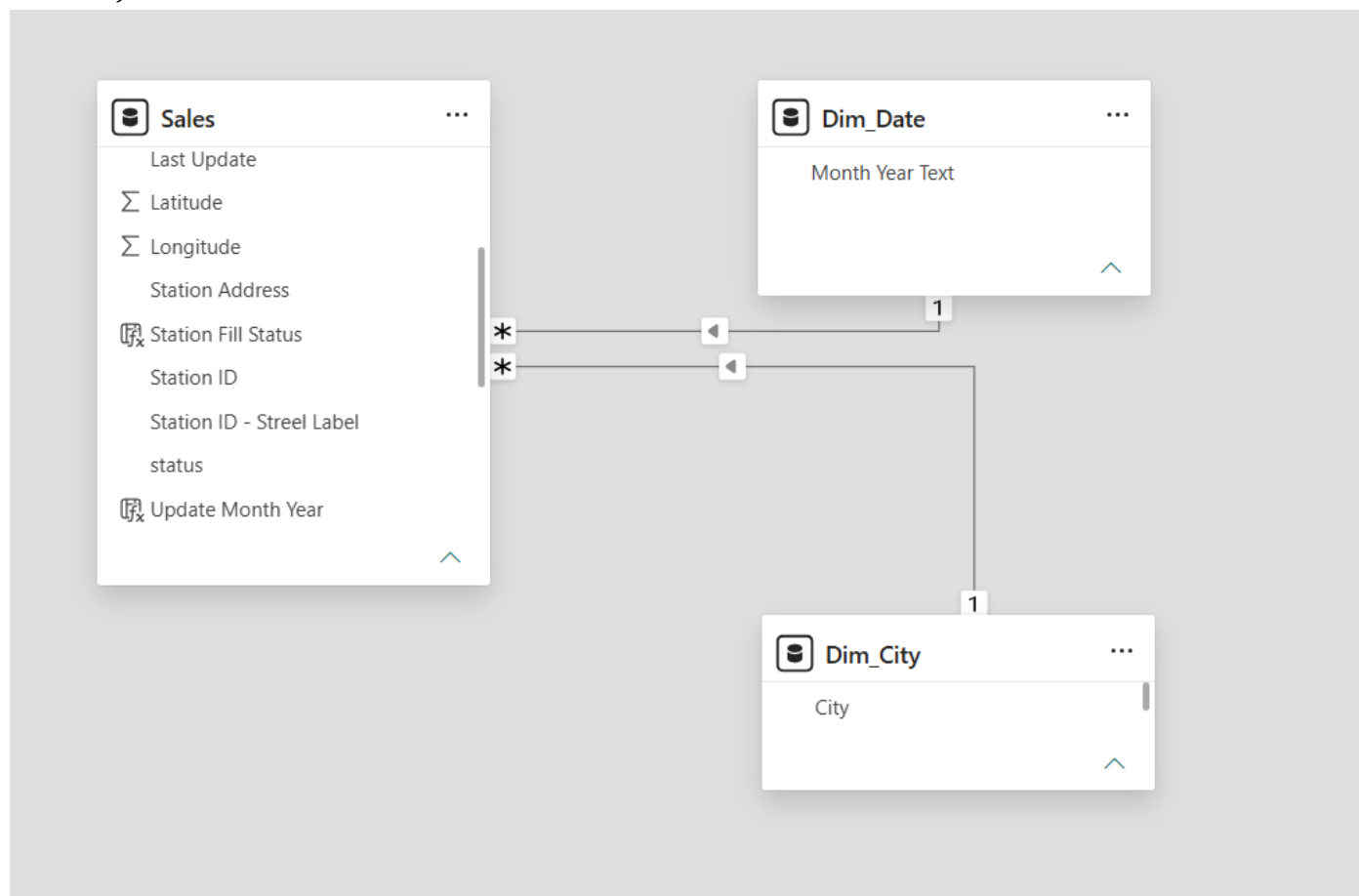
A star-schema model was built with one fact table and two dimension tables:

Fact Table

- Sales — station-level operational records (Bike Stands, Available Bikes, Available Bike Stands, status, banking, bonus, Last Update, Latitude, Longitude)

Dimension Tables

- Dim_City — unique list of 25 city contracts, related to Sales via Contract Name (Many-to-1)
- Dim_Date — unique list of Month-Year periods, related to Sales via Update Month Year (Many-to-1)



8. DAX Measures

16 DAX measures were created to support the dashboard. Key examples:

Missing Bike Capacity =
SUMX('Sales', VALUE('Sales'[Bike Stands]) -
(VALUE('Sales'[Available Bikes]) + VALUE('Sales'[Available Bike Stands])))

Active Bike Stations =
CALCULATE(
SUMX('Sales', VALUE('Sales'[Bike Stands])),
'Sales'[status] = "OPEN"
)

```
Utilization Rate =  
DIVIDE(  
SUMX('Sales', VALUE('Sales'[Bike Stands])) - SUMX('Sales', VALUE('Sales'[Available Bikes])),  
SUMX('Sales', VALUE('Sales'[Bike Stands]))  
)  
  
Bonus Enabled Stations =  
CALCULATE(COUNTROWS('Sales'), 'Sales'[bonus] = "TRUE")  
  
Banking Enabled Capacity =  
CALCULATE(  
SUMX('Sales', VALUE('Sales'[Bike Stands])),  
'Sales'[banking] = "TRUE"  
)  
  
Total Updates 2025 =  
CALCULATE(  
COUNTROWS('Sales'),  
YEAR(DATEVALUE(LEFT('Sales'[Last Update], 10))) = 2025  
)
```

Full measure list: Missing Bike Capacity, Total Bike Stands, Total Available Bikes, Total Available Bike Stands, Active Bike Stations, Inactive Bike Stations, Bonus Enabled Stations, Bonus Disabled Stations, Banking Enabled Capacity, Banking Disabled Capacity, Total Contracts, Utilization Rate, and Total Updates for 2022, 2023, 2024, and 2025.

9. Dashboard Features

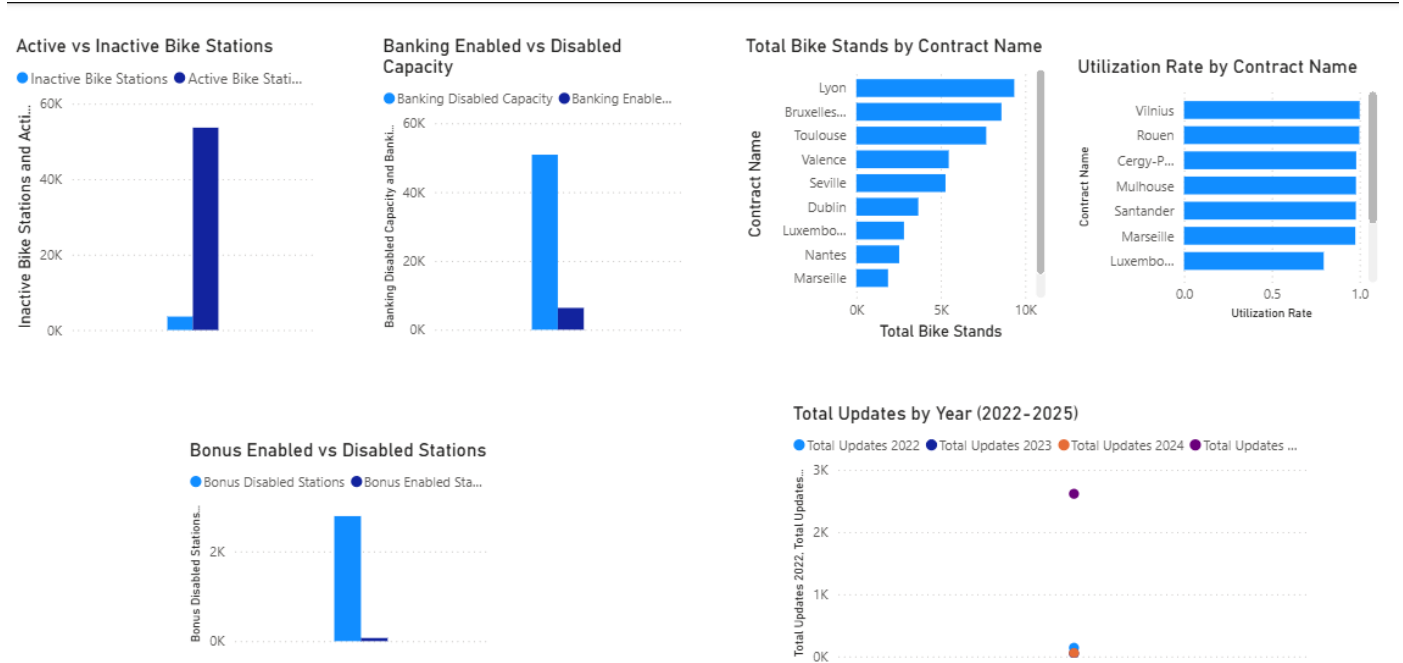
KPI Cards

- Total Bike Stands, Total Contracts, Total Available Bikes, Total Available Bike Stands, Missing Bike Capacity, Utilization Rate



Charts

- Active vs Inactive Bike Stations (clustered column)
- Banking Enabled vs Disabled Capacity (clustered column)
- Bonus Enabled vs Disabled Stations (clustered column)
- Total Updates by Year, 2022-2025 (line chart)
- Total Bike Stands by City — Top 10 (horizontal bar)
- Utilization Rate by City — Top 10 (horizontal bar)
- Station locations map, colored by status (OPEN/CLOSED)



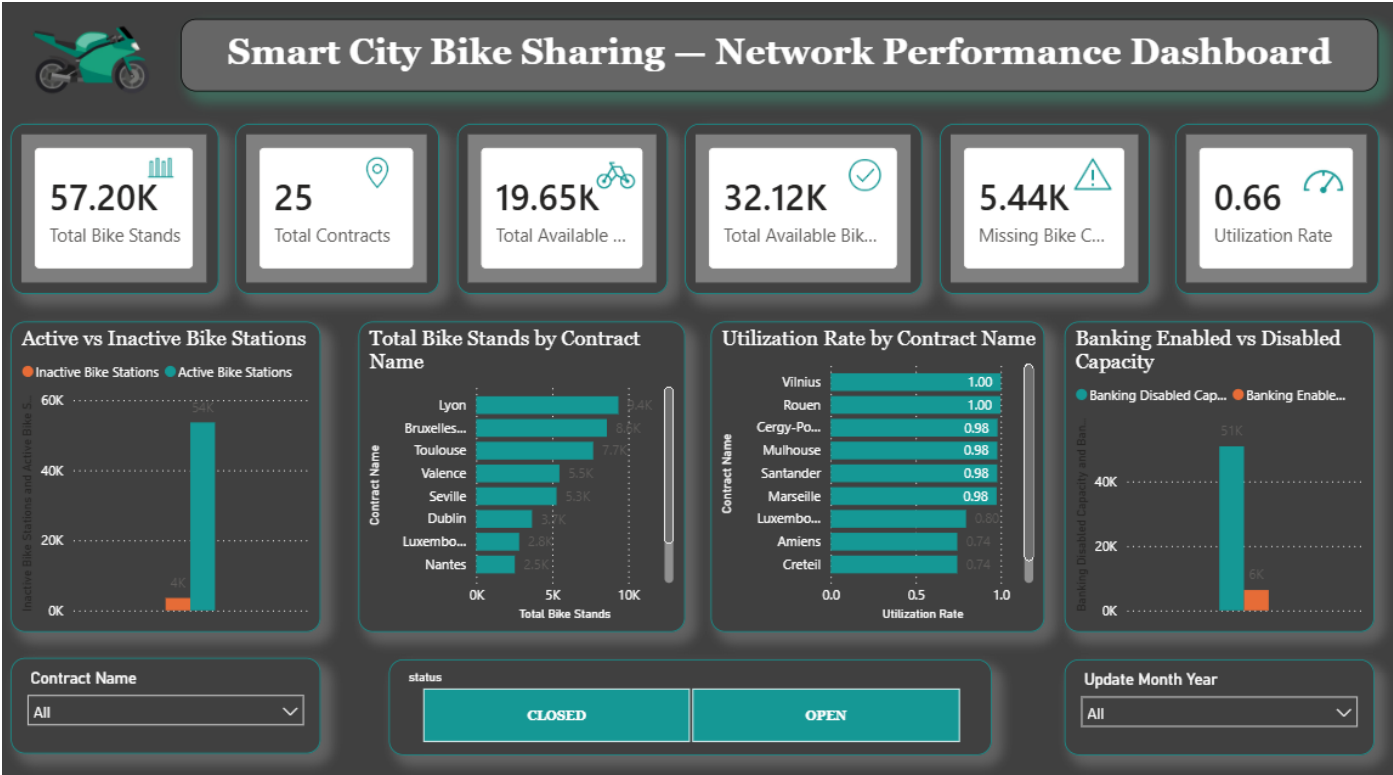
Filters / Slicers

- Contract Name (City)
- Status (OPEN / CLOSED)
- Update Month Year

Final Dashboard

- Title bar with bicycle icon
- 6 KPI cards: Total Bike Stands (57.20K), Total Contracts (25), Total Available Bikes (19.65K), Total Available Bike Stands (32.12K), Missing Bike Capacity (5.44K), Utilization Rate (0.66)
- 4 charts: Active vs Inactive Bike Stations, Total Bike Stands by Contract Name (Top 10), Utilization Rate by Contract Name (Top 10), Banking Enabled vs Disabled Capacity
- Slicers: Contract Name, Status (OPEN/CLOSED), Update Month Year
- 4 more KPI cards: Total Updates 2025 (3K), 2024 (55), 2023 (47), 2022 (139)
- Station Locations map, colored by status
- Bonus Enabled vs Disabled Stations chart

- Inactive Bike Stations card (3.56K)
- Total Updates by Year (2022-2025) chart



10. Insights and Conclusions

Total Bike Stand Capacity	57,203
Total Available Bikes	19,646
Total Available Bike Stands	32,118
Missing / Mismatched Capacity	5,439 (9.5%)
Active Station Capacity (OPEN)	53,650 (93.8%)
Inactive Station Capacity (CLOSED)	3,560 (6.2%)
Bonus-Enabled Stations	70 of 2,856
Banking-Enabled Capacity	6,330 (11.1%)
Total Contracts (Cities)	25
Overall Utilization Rate	66%
Total Records Analyzed	2,856
Data Span	18 Jan 2022 – 12 Nov 2025
Records in 2025	2,615 (91.6% of dataset)

Key Findings

Summarizing what the data shows, as-is.

- The network operates 57,203 total bike stands across 25 city contracts.
- 19,646 bikes were available for rent, and 32,118 docks were open for returns at the time of analysis.
- 93.8% of total capacity (53,650 stands) sits at OPEN stations; 6.2% (3,560 stands) is CLOSED.
- Only 70 stations (2.5%) offer bonus incentives; 6,330 stands (11.1%) support banking/card payment.
- Overall network utilization rate is 66%.
- 2,856 records were captured between 18 Jan 2022 and 12 Nov 2025, with 91.6% of all records dated in 2025 alone.

Digging into causes behind the patterns above.

- Why is there a 9.5% capacity mismatch (5,439 stands)? Likely causes: API synchronization delays between the sensor feed and the reporting system, bikes temporarily removed for maintenance, docking points under repair, or bikes in transit during redistribution — not a genuine loss of physical infrastructure.
- Why does 2025 dominate the dataset (91.6% of records)? This reflects a recent increase in data collection/refresh frequency (the API likely started polling more often or was only recently connected at scale), not a sudden four-year expansion of physical stations.
- Why is utilization only 66% rather than near 100%? A third of capacity sitting idle at any moment is expected and healthy in bike-share systems — it reflects riders returning bikes throughout the day rather than the system running at full capacity constantly; persistently low utilization at specific stations would be the more concerning signal.

- Why do so few stations offer bonus incentives (2.5%)? Bonus stations are typically placed at physically inconvenient locations (uphill, edge-of-network) specifically to encourage riders to return bikes there — a small proportion is expected by design, not a gap.

What current patterns suggest about the near future — framed as informed expectations, not guarantees, since this dataset is a snapshot rather than a time series with enough consistent history to model formally.

- If the 9.5% capacity mismatch pattern continues unaddressed, reported availability will keep diverging from real availability, which risks growing rider complaints about "phantom" bikes/docks that aren't actually there.
- Given data collection is clearly accelerating (91.6% of records from 2025 alone), future datasets will likely have far denser, more reliable time coverage, making genuine year-over-year trend analysis increasingly possible going forward.
- Cities with the highest Utilization Rate (Vilnius, Rouen, Cergy-Pontoise — all near 0.98-1.00) are likely to experience bike shortages during peak hours if capacity isn't increased, since they're already running close to fully utilized.
- Contracts with large total capacity but lower utilization may be early indicators of oversupply relative to demand in that city, worth monitoring before further station expansion there.

Concrete, actionable recommendations tied to the findings above.

- Implement an automated validation rule: flag any station where $\text{Bike Stands} \neq \text{Available Bikes} + \text{Available Bike Stands}$, and route it for manual/maintenance review rather than letting the 9.5% gap persist silently.
- Increase refresh frequency for historical-equivalent tracking going forward, so 2026 onward has consistent month-by-month density (avoiding another "2025-style" data concentration).
- Prioritize bike redistribution in high-utilization cities (Vilnius, Rouen, Cergy-Pontoise) during peak hours to prevent empty-station shortages.
- Expand banking-terminal coverage beyond the current 11.1% of capacity, particularly in high-utilization cities, to reduce dependency on app-only rentals.
- Schedule maintenance reviews for CLOSED stations (6.2% of total capacity) to bring more capacity back online.
- Reassess bonus-station placement in cities with heavy one-directional usage (e.g., downhill commuter routes) to help naturally rebalance bike distribution.

11. Conclusion

This project transformed a raw, 2,856-row bike-sharing station export into a structured, analyzable data model and an interactive Power BI dashboard. Through Power Query cleaning, star-schema modeling, and 16 DAX measures, the analysis surfaced genuine operational insights: a network operating at 93.8% availability, a 66% utilization rate, and a 9.5% capacity-reporting gap worth investigating. Rather than fabricating unavailable revenue figures, the analysis was scoped honestly to what the data could support — capacity, availability, and operational status — with recommendations focused on data quality and operational efficiency for the bike-sharing network.

Thank You!